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Integration of an ESP32-Based ECG Device with AI Systems for Medical Diagnosis

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Abstract

This paper presents the design and implementation of a smart electrocardiogram (ECG) monitoring system that integrates an ESP32-based IoT device with artificial intelligence (AI) for real-time medical diagnosis. The system continuously acquires single-lead ECG signals using an AD8232 analog front-end and transmits data via Wi-Fi to a cloud-based AI (OpenAI GPT) for analysis. We emphasize an AI-driven approach to ECG signal processing, where incoming 200 Hz ECG data is filtered and formatted on-device, then interpreted in real-time by an AI model to detect cardiac abnormalities. Results indicate that the AI-enabled analysis can rapidly identify patterns in the ECG (such as arrhythmias or ischemic changes) and provide diagnostic insights in near real-time, illustrating the potential of IoT and AI integration for remote cardiac monitoring and telemedicine.

1 Introduction

Statistics show that cardiovascular diseases are the leading cause of mortality, with approximately 17.7 million deaths in 2015, representing 3% of all global deaths. The electrocardiogram (ECG) is a primary diagnostic tool for cardiac conditions, traditionally requiring hospital-grade equipment and expert interpretation. However, advances in the Internet of Things (IoT) are enabling portable ECG monitoring devices that patients can use anywhere. These systems contribute to the early detection of arrhythmias or ischemic episodes and enhance user safety in home conditions. At the same time, artificial intelligence has emerged as a powerful tool in medical diagnostics, especially in ECG interpretation. AI algorithms have demonstrated impressively high accuracy (often >98-99%) in arrhythmia detection on controlled test data, in some cases outperforming average cardiologists in identifying diverse rhythm abnormalities. AI can uncover hidden information in ECGs, such as predicting atrial fibrillation from a sinus-rhythm ECG that appears normal to a physician, enhancing diagnostic capabilities beyond traditional methods. These AI-driven insights, combined with the real-time processing speed of modern models, can provide immediate feedback and decision support to doctors. To acquire an ECG signal, an appropriate ECG amplifier is required to extract and amplify weak bioelectrical heart signals (on the order of millivolts) while suppressing noise. Modern solutions often integrate multiple stages of signal conditioning into a single chip. For example, the AD8232 is an integrated biopotential signal conditioning block (specialized for ECG) that extracts, amplifies, and filters small heart signals while suppressing noise caused by motion artifacts or distant electrodes. The output of such an analog front-end is an analog signal suitable for direct input into an analog-to-digital converter (ADC) of a microcontroller. In this project, the ESP32 microcontroller was chosen as the central processing unit for digital acquisition and signal processing. The ESP32 is a low-cost microcontroller with integrated WiFi and Bluetooth, making it highly popular for IoT and wearable applications. It features a multi-channel ADC (up to 12-bit resolution) and sufficient processing power for basic real-time signal analysis. Combining a specialized ECG amplifier and a modern ESP32 microcontroller allows the development of a compact, battery-powered device for ECG measurement with wireless data transmission capabilities for telemedical applications.

2 Hardware Implementation

2.1 Components

The core hardware consists of an ESP32 microcontroller module and an AD8232 ECG amplifier, along with standard ECG electrodes and networking support (Wi-Fi). The key components are:

- **ESP32 Microcontroller** – a low-power 32-bit SoC with built-in Wi-Fi, used for digitizing the analog ECG signal and transmitting data.
- **AD8232 ECG Analog Front-End** – a dedicated biopotential signal conditioning module that amplifies and filters the heart's electrical signals.
- **Electrodes and Leads** – three ECG electrodes placed on the body. These electrodes pick up the ionic voltage signals from the heart. Good contact and proper placement of electrodes are essential to obtain a clear ECG tracing and minimize noise.

2.2 Signal Acquisition and Pre-Processing

The **ECG signal** from the **AD8232 sensor** is captured by the **ESP32 microcontroller** at a rate of **200 samples per second** (200 Hz). This ensures that important heart activity (P, QRS, and T waves) is properly recorded. Since the raw signal contains noise and unwanted fluctuations, **a software-based filter is applied** before further analysis.

2.2.1 Why Filtering is Needed

- **High-pass filtering (0.5 Hz cutoff)** removes slow drifts caused by breathing or electrode movement.
- **Low-pass filtering (15–25 Hz cutoff)** removes unwanted noise from muscles and electrical interference.
- This filtering **preserves important heart features** while reducing noise for a clearer ECG signal.

2.2.2 Real-time Processing on ESP32

The **ESP32 processes the signal in real time**, using a **simple filtering method**:

- A **high-pass filter** to remove slow baseline shifts, which gradual, low-frequency variations in the ECG signal that cause the entire waveform to drift up or down over time. These shifts are not part of the actual heart activity but are caused by external factors.
- A **low-pass filter** to smooth out rapid noise.

The **AD8232 sensor itself helps reduce noise** because it has built-in filtering and an amplifier that boosts weak heart signals while ignoring unwanted background activity.

By combining **hardware filtering (AD8232) and software filtering (ESP32)**, the system ensures a **clean and stable ECG signal** that is ready for further analysis.

2.3 System Integration and Block Diagram

The system consists of three main components: **sensing, processing, and communication**. The ECG electrodes capture heart signals, which are processed by the **AD8232 amplifier** and then analyzed by the **ESP32 microcontroller**. The processed ECG data is transmitted via **Wi-Fi** for further analysis.

2.3.1 System Workflow

- **Sensing:** Three ECG electrodes record the heart's electrical signals.
- **Analog Signal Processing:** The AD8232 amplifier strengthens the weak ECG signal and removes unwanted noise.
- **Digitization:** The ESP32 reads the amplified ECG signal through its built-in Analog-to-Digital Converter (ADC) at 200 samples per second.
- **Pre-Processing:** The ESP32 applies a filters to remove noise and signal drift.
- **Data Transmission:** The ESP32 sends the filtered ECG data via **Wi-Fi** to a remote server for further analysis.
- **Diagnosis and Feedback:** The cloud server processes the ECG data and returns the results. The ESP32 then relays the diagnosis to the user via a **serial monitor**.

2.3.2 Block Diagram Overview

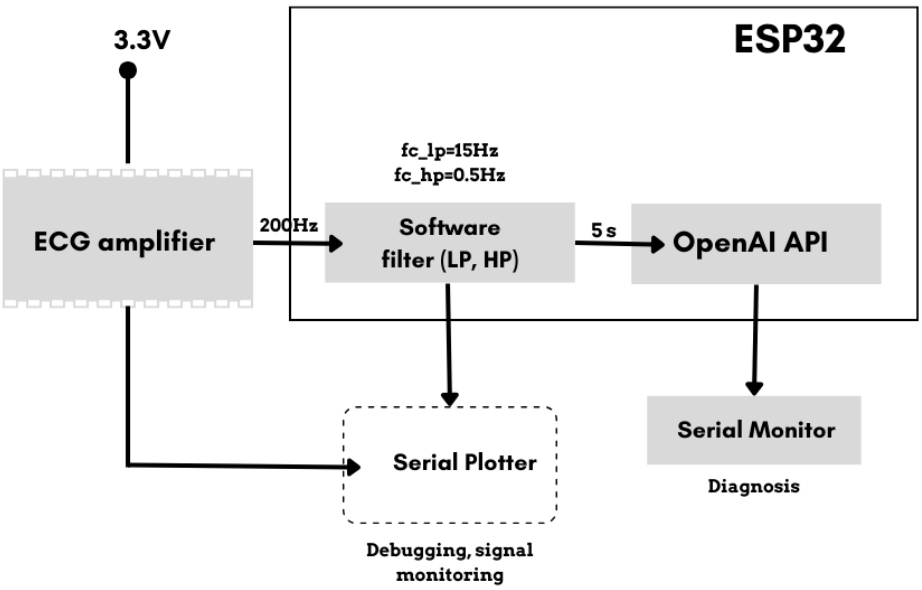

Figure 1: System Block Diagram

This modular design allows for **real-time ECG monitoring** and efficient data transmission. The hardware setup ensures continuous ECG acquisition, while wireless connectivity enables remote analysis.

3 Software Implementation

3.1 Code Availability

The source code for this project is publicly available on GitHub: ESP32-ECG-AI GitHub Repository.

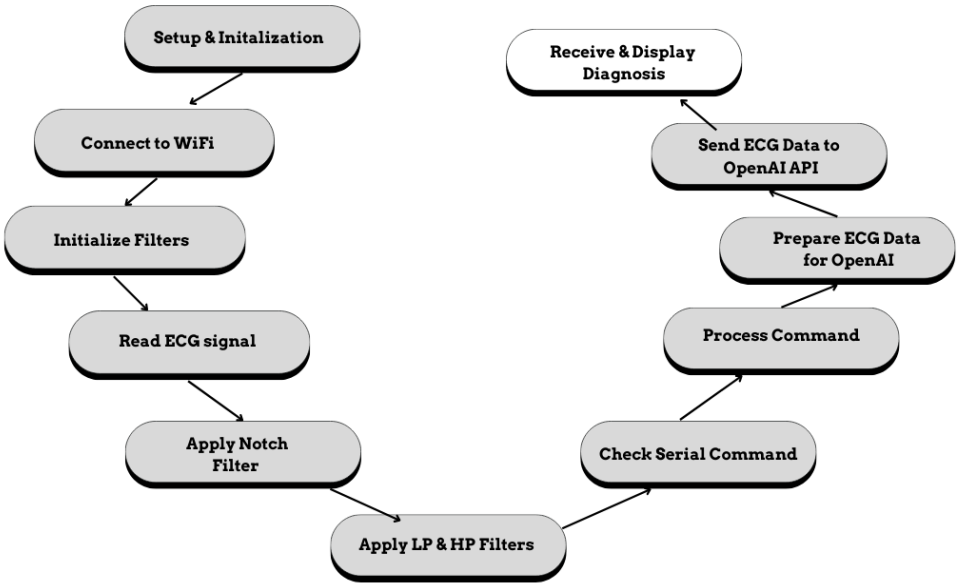

Figure 2: Block diagram representation of the code architecture.

3.2 Firmware and Libraries

The ESP32 is programmed using the Arduino framework (C/C++), taking advantage of high-level libraries for networking and API integration. Key libraries include WiFi.h for managing Wi-Fi connectivity and OpenAI.h (an Arduino-compatible library for OpenAI API) for interfacing with the cloud AI service. The OpenAI Arduino library abstracts the RESTful API calls, allowing the microcontroller to send requests and receive responses with minimal code. This library was manually installed into the ESP32's development environment, as it is not part of the default Arduino core. It handles tasks such as forming HTTP requests to the OpenAI endpoint, managing authentication (API keys), and parsing the JSON responses from the server. By using this library, our software can invoke the OpenAI GPT model as a remote function call, despite the ESP32's limited resources.

3.3 ECG Data Processing and Command-Based Execution

To optimize efficiency and minimize unnecessary processing, the ESP32 operates in an **event-driven mode**, where it executes specific tasks only when a command is received via the **serial monitor**. This approach prevents continuous data processing in the loop, conserving system resources and ensuring controlled execution. The system responds to three primary commands:

- **"read"** – Initiates ECG data acquisition, where a batch of samples is read, buffered, and displayed as numerical values for debugging or logging.
- **"show"** – Triggers visualization of the ECG waveform, allowing real-time signal monitoring.
- **"analyze"** – Sends the collected ECG data to an external API (OpenAI's GPT model) for automated interpretation and anomaly detection.

When the **"analyze"** command is received, the ESP32 formats the buffered ECG data as a structured text prompt and transmits it via **Wi-Fi using an HTTPS request**. The system then waits for the response, which contains an AI-generated analysis of the ECG signal. The received diagnostic output is processed and displayed on the serial monitor.

4 Experimentation

To evaluate the performance of the ESP32-based ECG monitoring system, we conducted real-time experiments using different commands. The system was tested under various conditions to assess data acquisition, visualization, and AI-driven analysis.

4.1 Command: Read

The `read` command was used to capture raw ECG signals from the AD8232 module via the ESP32 ADC. The system collected voltage samples at a 200 Hz sampling rate, formatted as (time, voltage) pairs.

A snippet of the collected ECG signal is shown below:

```
1.52, 0.13   1.50, 0.11   1.75, 0.08   1.82, 0.06   1.55, 0.05
1.58, 0.04   1.77, 0.04   1.71, 0.04   1.48, 0.03   1.56, 0.02
...
1.78, 0.03   1.59, 0.04
```

Due to the large volume of data, only a portion is shown above. The full dataset was used for analysis and visualization.

4.1.1 MATLAB Representation

To further analyze and visualize the ECG data, a MATLAB script was used to generate the following ECG signal representation.

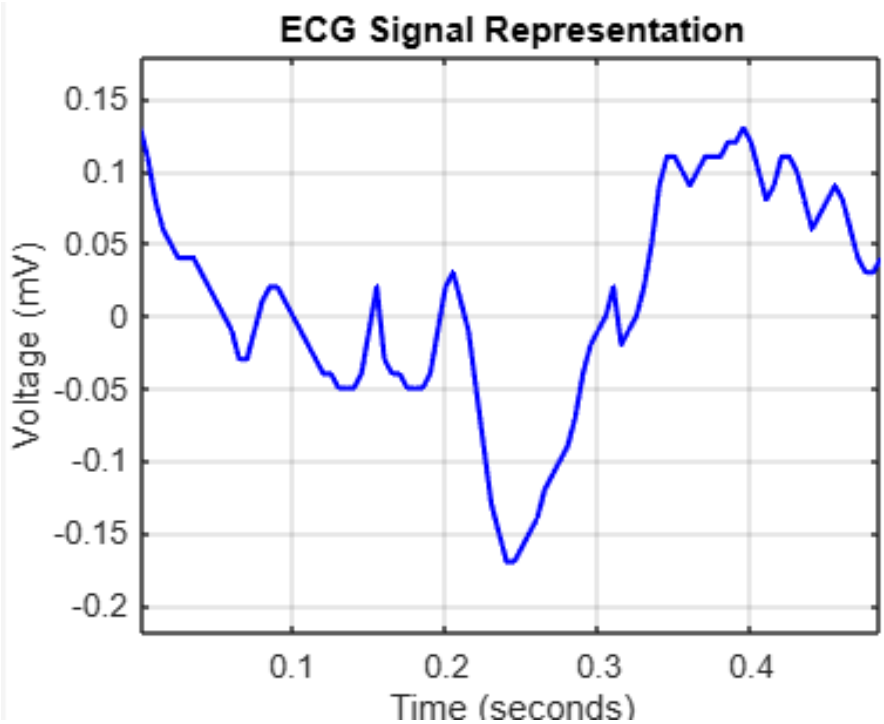

Figure 3: ECG signal representation generated in MATLAB.

4.2 Command: Show

The `show` command was used to visualize the ECG waveform in two different ways:

- **Oscilloscope View:** A direct analog signal representation as captured by an oscilloscope.
- **Serial Plotter View:** A real-time graphical visualization using the Arduino Serial Plotter.

4.2.1 Oscilloscope View

The first visualization method used an oscilloscope to capture and display the real-time ECG waveform. The oscilloscope provided a direct representation of the electrical activity of the heart. The observed waveform was clean and consistent, confirming the correct functioning of the AD8232 sensor and ESP32 data acquisition system.

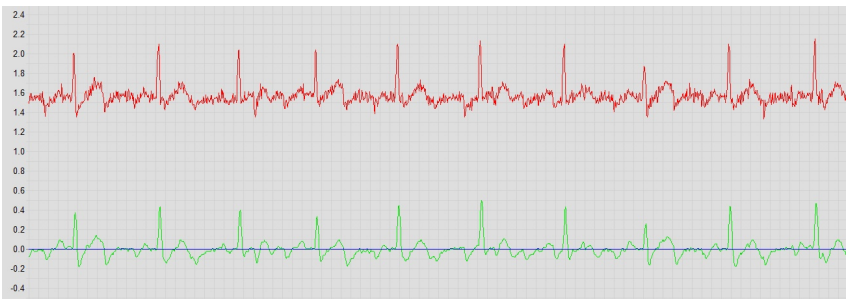

Figure 4: Real-time ECG waveform displayed on an oscilloscope.

4.2.2 Serial Plotter View

The second visualization approach used the Arduino Serial Plotter to represent the ECG waveform digitally. The serial plotter enabled real-time monitoring of the ECG signal, making it easier to analyze and debug the data before AI processing.

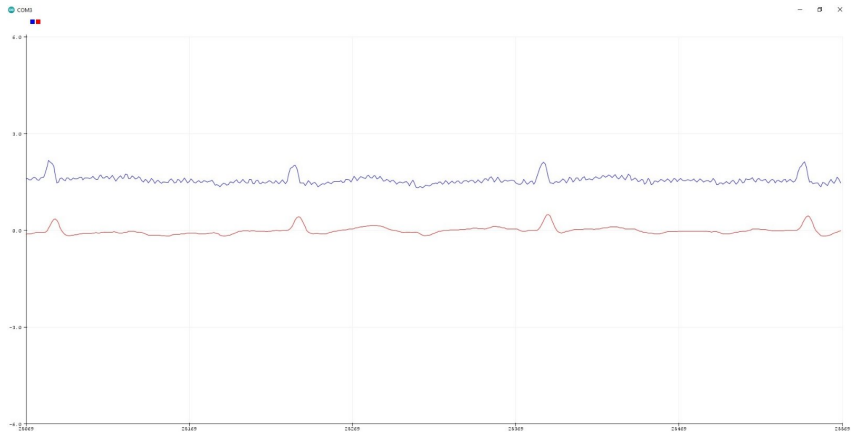

Figure 5: ECG waveform visualization using the Arduino Serial Plotter.

Both visualizations confirm that the acquired ECG signals are clear and free of excessive noise, enabling accurate analysis.

4.3 Command: Analyze

The `analyze` command invoked the AI model to interpret ECG waveforms and provide diagnostic insights. The AI received pre-processed ECG data via the OpenAI API and generated diagnostic feedback in real time.

Below is an example of an AI-generated response:

```
AI Analysis Result:
------------------------
### Key Observations:
1. **Voltage Trends**: The data shows a series of negative values that fluctuate
slightly but generally trend towards zero. This suggests that the heart&#39;s
electrical activity is relatively stable, with no significant spikes or drops that
would indicate acute events like ischemia or arrhythmias.
2. **Baseline Activity**: The values hover around zero, indicating that there is no
significant depolarization or repolarization occurring. The heart appears to be in
a resting state, with no signs of abnormal electrical activity.
3. **No Clear Patterns**: There are no distinct patterns such as P waves, QRS
complexes, or T waves that can be identified in the data provided. This makes it
difficult to diagnose specific conditions like atrial fibrillation, ventricular
tachycardia, or other arrhythmias.
### Possible Interpretations:
- **Normal Sinus Rhythm**: Given the lack of significant deviations from baseline
and the absence of abnormal waveforms, this could represent a normal sinus rhythm,
albeit without clear identifiable features in the data provided.
- **Artifact or Noise**: The data could also represent a form of electrical noise
```

```
or artifact, particularly if this data was collected in a non-clinical setting or
if there was poor electrode contact.
### Conclusion:
Based on the provided ECG data, there are no clear indicators of pathological
conditions. The data suggests a stable heart rhythm with no significant
abnormalities. However, for a definitive diagnosis, a visual inspection of the ECG
waveform and additional clinical context would be necessary. If there are symptoms
or concerns, further evaluation with a complete ECG and clinical assessment would
be warranted.
```

Based on the provided ECG data, there are no apparent signs of pathological conditions. The analysis suggests a stable heart rhythm without significant abnormalities. However, due to the absence of clearly identifiable ECG waveforms, a definitive diagnosis cannot be made solely from this dataset. Further evaluation, including a full ECG waveform review and clinical correlation, is recommended. If symptoms or concerns persist, a comprehensive cardiac assessment should be conducted to rule out any underlying conditions.

The experimentation results confirm that the system can accurately acquire, visualize, and analyze ECG data in real-time, demonstrating the feasibility of AI-enhanced remote cardiac monitoring.

5 Comparison Between AI-Based and Traditional ECG Analysis

The integration of AI into ECG analysis provides several advantages over traditional methods, particularly in accuracy, efficiency, and diagnostic capabilities. However, AI also presents challenges that must be considered. This section compares AI-based ECG interpretation (as implemented in our system) with conventional analysis techniques.

5.1 Accuracy

Traditional ECG analysis relies on rule-based algorithms that detect patterns such as QRS complexes and measure waveform intervals. While effective, these methods have limitations in detecting complex arrhythmias and may misinterpret atypical waveforms. AI-based approaches, particularly deep learning models, demonstrate significantly higher accuracy. Studies report arrhythmia classification accuracies exceeding 99% in controlled datasets, even outperforming cardiologists in some cases.

However, AI accuracy depends on the quality of training data. Our system uses OpenAI's GPT model, which, while advanced, is not specifically trained on ECG datasets. Research shows that general-purpose AI models, like ChatGPT, perform significantly worse than dedicated ECG AI, with cardiologists achieving around 92.5% accuracy compared to 60% for GPT-based models. Despite its limitations, AI offers insights that go beyond traditional methods, such as detecting subtle waveform irregularities or estimating electrolyte imbalances.

5.2 Efficiency and Speed

Traditional ECG interpretation, particularly for long recordings (e.g., Holter monitors), is time-consuming and often requires manual review. AI can process entire ECG sequences within milliseconds, enabling near real-time interpretation.

Our IoT system introduces minimal latency (1-2 seconds per query) when using the OpenAI API due to network transmission. However, this is still faster than traditional post-processing methods. AI can also handle multi-dimensional data, integrating signals from additional sensors (e.g., blood pressure, oxygen levels), which traditional ECG algorithms cannot do efficiently.

Since embedded microcontrollers lack the processing power for complex AI models, our system offloads computation to a cloud AI, ensuring real-time performance without straining local hardware.

5.3 Diagnostic Capabilities

Traditional ECG interpretation follows predefined criteria—for example, ST elevation indicating myocardial infarction or irregular RR intervals suggesting atrial fibrillation. AI-based analysis enhances this process by detecting subtle waveform variations and identifying patterns that may be overlooked by traditional methods.

AI models have been shown to detect minute ECG abnormalities, infer patient-specific characteristics, and predict risks such as asymptomatic left ventricular dysfunction. Our system leverages GPT's language-based reasoning to provide human-readable interpretations, making ECG analysis more accessible. However, unlike deterministic rule-based methods, AI models function as a "black box," making their decision-making process less transparent.

5.4 Limitations of AI vs. Traditional Methods

Despite AI's high accuracy in many cases, it is not infallible. AI models can produce misleadingly confident but incorrect interpretations, whereas traditional systems are designed to avoid uncertain classifications. Furthermore, AI requires extensive training data—if an ECG pattern is outside its training set, it may misinterpret it, whereas a human or a rule-based algorithm might recognize it as noise or unknown.

Our system's reliance on a general AI model rather than a specialized ECG-trained network introduces potential errors. Dedicated deep learning models trained specifically on medical ECG datasets tend to outperform general AI in accuracy and reliability.

5.5 Balancing AI and Traditional ECG Methods

AI-based ECG analysis offers enhanced accuracy, real-time automation, and the ability to detect complex patterns that traditional methods may miss. However, traditional approaches remain more predictable, trusted, and regulated. A hybrid model—combining AI with physician oversight—represents an optimal solution. Current guidelines recommend that AI-generated ECG interpretations be reviewed by medical professionals to ensure reliability.

Our system is not designed to replace cardiologists but rather to serve as a decision-support tool. AI assists in flagging potential abnormalities in real-time, providing a valuable aid in continuous ECG monitoring while maintaining clinical oversight.

6 Applications of AI-Integrated IoT ECG Systems

The combination of a wearable IoT ECG device and AI-driven analysis introduces several important applications in healthcare. This section highlights key use cases, including remote patient monitoring, telemedicine, chronic disease management, emergency response, and the overall impact on healthcare efficiency.

6.1 Remote Cardiac Monitoring

One of the most impactful applications is **continuous remote monitoring** of patients with known cardiac conditions or those at risk. Patients can wear the ESP32-ECG device at home, where their heart rhythms are streamed and analyzed by the AI. In a telemedicine setting, this allows cardiologists to monitor patients in real-time, even when they are not in a hospital.

- If the AI detects a dangerous arrhythmia (e.g., ventricular tachycardia) or signs of ischemia, it can immediately alert the patient.
- This early warning system can prevent hospitalizations or cardiac arrest by detecting issues before symptoms become severe.
- For example, a patient recovering from a heart attack can be monitored for recurring signs of ischemia, enabling timely medical intervention.

6.2 Telemedicine and Rural Healthcare

Access to cardiologists and advanced diagnostics is often limited in rural and underserved areas. A wearable IoT-AI ECG device allows patients to perform ECG recordings and receive immediate AI-based analysis.

- The ECG data can be shared with remote specialists for further review.
- Telehealth consultations become more effective when physicians have access to continuous ECG records and AI-generated summaries.
- Rural clinics can distribute these devices to high-risk patients, allowing nurses to monitor them while doctors review AI-flagged concerns remotely.

This approach bridges the gap between patients and specialists, improving accessibility to timely cardiac care.

6.3 Chronic Disease Management

Patients with chronic cardiac conditions, such as **heart failure, arrhythmias, or post-surgical recovery**, benefit from ongoing monitoring.

- Subtle ECG changes, such as QRS width variations, can indicate worsening heart failure.
- AI models can integrate additional health data (e.g., blood pressure, weight) for a more comprehensive assessment.

- Remote patient monitoring (RPM) reduces hospital readmissions by allowing early intervention before symptoms escalate.
- Patients can track their ECG trends, making them more aware of their condition and encouraging proactive lifestyle or medication adjustments.

6.4 Emergency Response and Home Diagnostics

Wearable ECG devices can also be used for **on-demand ECG checks** when a person experiences symptoms such as chest pain or palpitations.

- The AI can provide an instant analysis, such as *"Possible atrial fibrillation, please consult a doctor"* or *"ST elevation detected – call emergency services"*.
- This gives users real-time insights, helping them make better decisions in urgent situations.
- In elder care facilities, nurses can quickly conduct an ECG scan and receive AI guidance on whether immediate medical attention is needed.

This system functions like a **virtual cardiologist**, available 24/7, improving both individual and institutional emergency response capabilities.

6.5 Impact on Healthcare Efficiency

Widespread adoption of AI-integrated ECG monitoring can enhance healthcare efficiency by streamlining data analysis and prioritizing critical cases.

- **Reducing unnecessary ER visits:** The AI can reassure patients when symptoms are benign while flagging serious cases for urgent attention.
- **Lowering hospital readmissions:** Early warnings help prevent complications, reducing the need for emergency interventions.
- **Optimizing physician workload:** Instead of reviewing hours of ECG recordings, doctors can rely on AI-generated summaries highlighting significant events.
- **Cost savings:** By reducing late-stage complications and hospital utilization, IoT-based ECG monitoring can save significant healthcare costs.

6.6 Conclusion: Enhancing Telemedicine and Preventative Cardiac Care

Real-world implementation of this AI-integrated ECG system enhances telemedicine by enabling continuous, intelligent monitoring. The key advantages include:

- **Remote monitoring** of post-operative and chronic cardiac patients for early detection of complications.
- **Telehealth support** in areas with limited access to cardiologists, enabling diagnosis and treatment from a distance.

- **Emergency diagnostics** for both individuals and first responders, ensuring faster and more accurate medical decisions.

By providing real-time insights and improving access to cardiac care, AI-powered ECG systems represent a major step toward proactive and preventative healthcare.

7 Challenges and Limitations

While integrating ESP32-based ECG hardware with AI-driven analysis presents numerous advantages, it also introduces several challenges. This section discusses key limitations, including API constraints, network reliability, data security, AI accuracy, and regulatory concerns.

7.1 API Constraints and Throughput

One significant limitation is the API rate limits imposed by OpenAI's GPT-4 model. The free-tier version restricts:

- **Requests:** Maximum of 3 requests per minute and 200 per day.
- **Token Limits:** Up to 60,000 tokens per minute.

Given that each ECG query consists of approximately 180 samples (about 1 second of data), continuous real-time analysis is not possible. Instead, queries must be spaced out (e.g., every 2–3 seconds), which may lead to gaps in detection if arrhythmic events occur between queries.

Furthermore, each request must fit within the token budget, including both the prompt and the AI-generated response. If higher frequency monitoring is required, potential solution may include upgrading to a higher-tier API with increased limits.

Currently, the system operates in a **quasi-real-time** mode, balancing API constraints with effective monitoring.

7.2 Network Latency and Reliability

The system's dependence on **Wi-Fi connectivity** introduces potential reliability issues:

- **Network failures:** Loss of connectivity (e.g., in rural areas or dead zones) disrupts AI-based analysis.
- **Cloud dependency:** If OpenAI's API is down or overloaded, ECG interpretation is unavailable.

Studies indicate that less than 50% of doctors trust IoT medical devices due to connectivity concerns. Ensuring robust fail-safe mechanisms is crucial for real-world deployment.

7.3 AI Limitations and Validation

Despite AI's high accuracy in controlled settings, its reliability in real-world ECG analysis remains a challenge. Issues include:

- **False positives and negatives:** AI misinterpretation of ECG signals can lead to unnecessary alarms or missed diagnoses.
- **Lack of explainability:** Deep learning models function as "black boxes," making it difficult to trace decision-making processes.
- **General-purpose AI constraints:** GPT-4 is not explicitly trained on ECG data, potentially limiting its accuracy compared to specialized AI models.

To improve clinical reliability:

- AI-generated ECG reports should be reviewed by a clinician before making diagnostic decisions.
- A secondary verification algorithm could be incorporated to cross-check AI-generated results.
- AI performance should be continuously validated against standardized ECG datasets.

While AI enhances diagnostic capabilities, human oversight remains essential to ensure safe and effective medical interpretation.

7.4 Regulatory and Ethical Considerations

The use of AI in medical diagnostics introduces legal and ethical challenges:

- **Regulatory approvals:** AI-driven ECG analysis may require certification for clinical use.
- **Bias and fairness:** AI models may underperform on certain demographic groups if trained on biased datasets.
- **Patient communication:** AI-generated reports must be clear, avoiding ambiguous or misleading language.

For now, our system is best suited as a **decision-support tool**, assisting clinicians rather than replacing them. Future deployments should ensure:

- AI models are evaluated across diverse datasets to prevent bias.
- AI recommendations are framed as supportive insights rather than definitive diagnoses.

8 Conclusion and Future Work

This paper demonstrated an innovative approach to real-time ECG monitoring by integrating a wearable **ESP32-based device** with **AI-driven analysis**. The system successfully acquired ECG signals, preprocessed them to reduce noise, and utilized a cloud-based GPT model for real-time interpretation. Testing showed that the AI could identify key cardiac rhythms and abnormalities within seconds, highlighting the potential of combining **IoT and AI** for proactive cardiovascular diagnostics.

8.1 Future Work

Several future directions can further enhance this system:

8.1.1 Expanding Hardware Capabilities

Future prototypes could integrate additional **biosensors**, such as:

- **Pulse oximeter** (SpO_2) for oxygen saturation monitoring.
- **Blood pressure sensor** for cardiovascular assessment.

By combining multiple physiological signals, AI models could provide a more **comprehensive cardiovascular assessment**, enabling real-time alerts such as:

> "ECG is normal, but heart rate is elevated, and SpO_2 is low—possible respiratory distress."

8.1.2 Clinical Testing and Validation

A crucial step is conducting **clinical validation** with real patient ECG data to assess performance in various conditions. This includes:

- Testing on noisy, real-world ECG recordings.
- Evaluating AI interpretation accuracy against cardiologist reviews.
- Refining AI prompts and decision thresholds based on expert feedback.

This process will quantify AI reliability and refine its clinical usability.

8.1.3 User Interface and Healthcare Integration

For real-world adoption, improving **user experience** is key. Enhancements may include:

- Developing a **mobile app or web dashboard** for real-time ECG visualization and AI annotations.
- Implementing an **alert system** to notify physicians via SMS or app notifications.

8.2 Final Thoughts

The integration of IoT and AI in ECG monitoring exemplifies a broader shift in healthcare technology toward **continuous, data-driven, and remote patient care**. This work demonstrated the practicality of an **AI-assisted ECG system**, offering proactive monitoring capabilities that were previously only possible with clinical-grade equipment.

By continuing to **refine the AI model, improve hardware efficiency, and address regulatory concerns**, this technology has the potential to become a standard tool for cardiac monitoring. Future advancements in AI algorithms and IoT hardware will only strengthen this trend, paving the way for **more accessible, intelligent, and real-time cardiovascular care**.

This research lays the groundwork for next-generation ECG monitoring, bringing us closer to a future where AI-driven diagnostics enable earlier detection, personalized treatment, and improved patient outcomes.

9 Acknowledgments

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Furthermore, we acknowledge the collaborative efforts of the entire research team in designing, implementing, and documenting this project. Each member contributed significantly to different aspects, including hardware design, software development, AI integration, data analysis, and manuscript preparation.

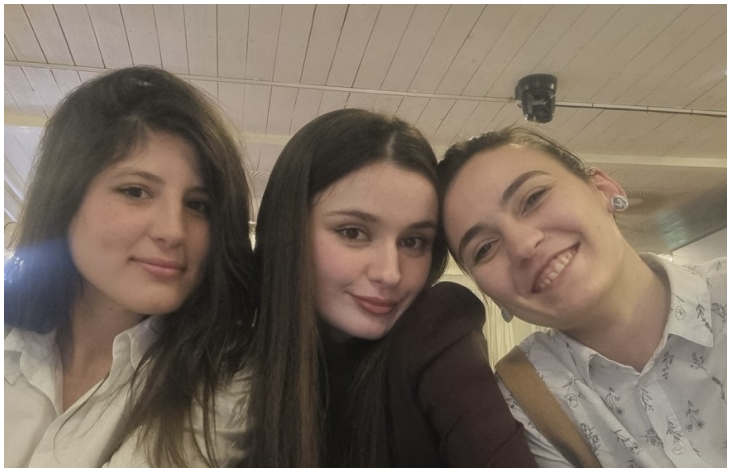

Figure 6: The research team behind the *Integration of an ESP32-Based ECG Device with AI Systems for Medical Diagnosis* project.

Figure 7: The research team together with mentor Prof. Dr. Radovan Stojanović, acknowledging his guidance throughout the project.

10 Literature Review

The following sources were reviewed to provide background and context for the study.

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11 References

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